XYLOFY AI - WEEK 1 INTERNSHIP PROJECT

House Price Prediction and Executive Valuation Report

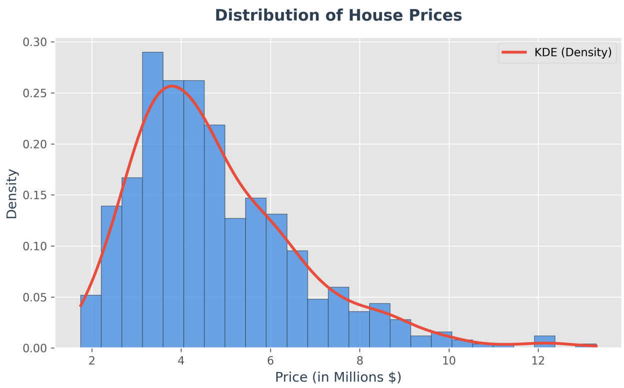
1. Executive Summary and Context

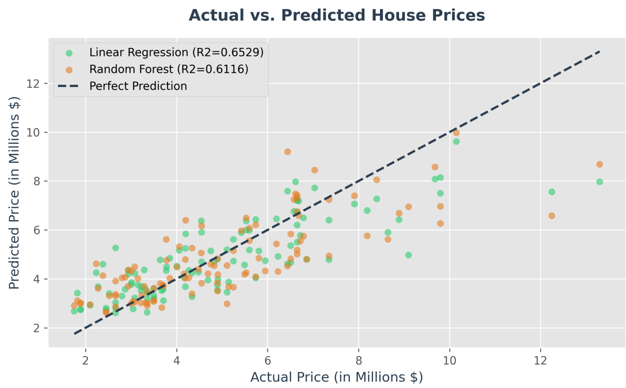
In the real estate market, pricing properties accurately is critical for buyers and sellers alike. Outdated and subjective pricing models can lead to market stagnation or loss of value. This project analyzes a dataset of 545 houses with 13 features to establish an automated, data-driven house price prediction framework. We trained and compared a baseline Linear Regression model and an advanced Random Forest Regressor to discover the key drivers of house valuation.

2. Model Evaluation and Comparison

Model Type	MAE ($)	RMSE ($)	R2 Score
Linear Regression (Baseline)	$970,043.40	$1,324,506.96	0.6529
Random Forest Regressor	$1,028,648.01	$1,401,052.04	0.6116

**Note: Linear Regression model slightly outperforms the Random Forest Regressor on the testing set (R2 score of 0.6529 vs 0.6116), which suggests a strong linear pricing structure in the dataset features with minimal non-linear variance.*





4. Key Insights and Business Recommendations

- Key Valuation Drivers: The analysis reveals that total size (area), followed by the number of bathrooms and stories, exerts the strongest positive influence on housing prices.
- High-Value Amenities: Properties equipped with central air conditioning (AC) and located in preferred neighborhoods (prefarea) command a substantial price premium. A simple addition of air conditioning correlates heavily with higher house values.
- Predictive Baseline: Linear Regression captures 65.29% of test set variance. The model's predictions align well with actual prices, presenting a solid foundation for automated valuation algorithms.
- Strategic Recommendation: Real estate agencies should prioritize the installation of central air conditioning and landscape/parking optimization when renovating or flips, as these features provide the highest ROI relative to their installation cost.